Phantasos-Lite — Cross-Experiment Synthesis

Synthesizes Experiments 1–3 (Qwen2.5-7B-Instruct, Qwen2.5-1.5B-Instruct, Llama-3.2-3B-Instruct) into the methodology, results, interpretation, rigor, limitations, and related-work material needed for the paper's Methods, Results, Discussion, and Conclusion sections. Built from the three per-experiment logs; no new numbers here beyond what those logs already report.

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1. Methodology

The same pipeline, built and validated once on the 7B model, was re-run on each subsequent model — not copy-pasted blindly, but with every model-specific mechanic (tokenizer, chat template, hook behavior) reverified fresh each time rather than assumed to transfer. Steps marked ⭐ are the ones doing load-bearing control work, not just procedural bookkeeping — each is explained inline.

Step 0 — Environment and extraction-mechanics verification (rerun fresh per model)

- Load the model in bf16, no quantization. ⭐ Quantization changes the actual activation values being studied, so this is a fidelity control, not a performance choice.
- Hook-output-type check — confirms what a forward hook on the model's decoder layer actually returns (tensor vs. tuple) before any extraction code assumes a shape.
- Digit-token-id discovery, with every digit asserted single-token. ⭐ This isn't incidental — the entire self-rating channel's `digit_mass`/expected-value readout is only valid if every counted token id genuinely corresponds to a complete digit, not a fragment.
- Chat-template-tail check — confirms the formatted prompt ends exactly at the generation point, with nothing trailing.
- Hook-correctness checks: **alpha=0 no-op** ⭐ (proves the steering hook is a true no-op until deliberately activated — without this, no extracted activation in the entire study could be trusted) and **per-position delta norm** ⭐ (proves position-targeting is exact, not leaking into neighboring sequence positions).

Step 1 — Persona and prompt construction (held constant across all three models)

- The same 16 personas, same taxonomy (anchor / expert / warm / cold / playful clusters), and the same 4 deliberately-confusable "surprise" pairs are used in every experiment. ⭐ Holding the test material fixed is what makes a cross-model confusion-pattern comparison meaningful — a difference in S5 confusions reflects the model being studied, not a shifting test.
- 2 independent system-prompt variants per persona. ⭐ This is what enables both the leave-one-variant-out S5 accuracy check and the split-half reliability / noise-ceiling

computation — without a second variant, there would be no way to distinguish a model's genuine representational reliability from a single noisy sample.
- 8 elicitation prompts (I-set) used identically across models.

Step 2 — S2: persona expression (generation + independent judge)

- Descriptions generated per persona × variant × prompt (sampled, not greedy).
- Scored by an independent judge model (Groq-hosted), not the model under study. ⭐ Avoids the model grading its own persona-following, which would be circular.
- A manual eyeball check of raw generations is run for every experiment before trusting any downstream number. ⭐ This is what would catch a persona whose system prompt isn't giving the model enough to work with — a prompt-design failure that could otherwise masquerade as a "the representation lacks this distinction" finding.

Step 3 — Activation extraction

- One forward pass per (persona, variant, prompt) combination; last-token hidden state captured at **3 layers fixed in advance of seeing any result** — held at the same relative indices (`[7, 13, 21]`) across all three models for direct comparability. ⭐ Pre-registering the layers rather than selecting them after seeing which ones look best is what keeps this an honest correlational test rather than a post-hoc-cherry-picked one.
- Checkpointed and desync-guarded — a job is only considered "done" if its vectors are actually present in the saved store, not just listed in an index file. ⭐ This caught a real interruption (Experiment 3's mid-run disconnect) and a real layer-mismatch rerun (Experiment 2) without either silently analyzing missing or stale data.

Step 4 — RDM construction (four independent channels)

- **RDM-A**: activation geometry (the thing being explained)
- **RDM-P**: embedding-based text baseline (the confound being controlled for)
- **RDM-F**: self-description, judge-scored free text
- **RDM-S**: self-rating, judge-free logit-expected-value digit readout (steering strength alpha=0 throughout — read-only, not an intervention)
- Diagonal self-ratings included in RDM-S as a validity spot-check. ⭐

Step 5 — Statistical analysis

- RSA (Spearman correlation) between RDM-A and each of RDM-P / RDM-F / RDM-S, per layer.
- **Permutation test** against a label-shuffled null, for every reported RSA value. ⭐ Establishes significance against a proper null distribution, not just "RSA is greater than zero."

- **Shuffled-label control**, run as a separate sanity check beyond the permutation test itself. ⭐ Verifies the RSA computation's implementation isn't trivially inflated by something other than genuine structure.
- **Partial RSA**, controlling for RDM-P. ⭐ The single most important control in the whole pipeline — without it, "personas correspond to activations" could just mean "personas use different words, and different words produce different activations," which would be a much less interesting finding.
- **Split-half reliability / noise ceiling** (variant 0 vs. variant 1 of RDM-A). ⭐ Makes it possible to tell a genuinely weak correspondence apart from a noisy measurement of a possibly strong one — the basis for the ceiling-normalized cross-model comparison in §3.

Step 6 — Cross-model replication

- The identical pipeline, persona set, and relative layer indices, rerun on each new model with every model-specific mechanic (Step 0) reverified from scratch rather than inherited. ⭐ This is what makes Experiments 2 and 3 a true replication of Experiment 1 rather than three independently-interpretable single-model studies — the only things that changed between runs were the model itself and what Step 0 discovered needed re-checking.

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2. Experiment setup — what to note honestly, per model

Experiment 1 — Qwen2.5-7B-Instruct. The original run; predates this round of live debugging, so its numbers are taken as reported in its own results log rather than re-derived here. Serves as the fixed reference point both later experiments were deliberately matched against (same persona set, same relative layer indices).

Experiment 2 — Qwen2.5-1.5B-Instruct. `PRIMARY_LAYERS` was initially computed as `round(n_layers * f)` for depth-fractions (0.25, 0.5, 0.75), yielding `[7, 14, 21]` — which silently diverged from Experiment 1's hand-picked `[7, 13, 21]` (13/28 is 46.4% depth, not exactly 50%). This was caught by auditing the figures against the printed numbers before it entered any cross-model table, not during the run itself, and corrected by hardcoding `PRIMARY_LAYERS = [7, 13, 21]` and fully re-extracting. The layers unaffected by the fix (7 and 21) reproduced bit-for-bit identical results before and after — a useful confirmation that the extraction pipeline is deterministic given the same inputs, not a source of run-to-run noise. Honest weak point: the self-rating channel's EV calibration was notably worse than Experiment 1's (mean 7.09 vs. target 9, vs. 7B's 8.87), and S5 confusions were diffuse rather than landing on the designed pairs — both flagged as genuine signal-quality degradation at this scale, not hidden or averaged away.

Experiment 3 — Llama-3.2-3B-Instruct. A gated model, requiring license acceptance on HuggingFace before loading. A different tokenizer and chat template meant digit-token ids,

the chat-template tail, and all Step 0 hook checks were independently rediscovered and reverified rather than assumed from either Qwen run — and this caught a real, structural difference: Llama's default template injects a fixed, persona-independent date preamble into every system prompt, absent in Qwen's template (assessed low-risk for RDM structure since it is identical across all personas, and left unmodified rather than hand-patched). The self-rating channel showed a serious data-quality issue (66% of ratings below the pre-registered confidence threshold), which was diagnosed live via direct top-token probing rather than left unexplained, and attributed to a genuine model-specific weakness in terse single-digit instruction-following under this model's own chat-template structure — not a pipeline defect, and not fixable without deviating from the model's native formatting. A mid-run infrastructure interruption (the runner's machine disconnecting) was recovered cleanly via the same checkpoint-resume design validated in Experiment 2, losing no completed work.

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3. Clear results

The primary correspondence replicates in all three models, across two independently-trained model families, across a parameter range from 1.5B to 7B.

	7B (Qwen)	1.5B (Qwen)	3B (Llama)
Self-desc RSA (peak layer)	0.725 (L21)	0.672 (L21)	0.817 (L13)
Self-rating RSA (peak layer)	0.779 (L21)	0.687 (L21)	0.717 (L13)
Partial-out-text, peak layer	0.683 / 0.743	0.610 / 0.652	0.777 / 0.678
Perm-test p-value	⩽0.0008	⩽0.0008	⩽0.0004
Shuffled-label control	−0.047	−0.090	−0.063

Every model shows a statistically significant, non-trivial correspondence between what the model says about its persona (both free-text description and forced-digit self-rating) and the geometry of its own internal activations — and that correspondence survives partialing out the confound of prompt-text wording alone.

Two secondary results are consistent and clear:

- **Self-description is the more robust channel** — strong and well-behaved in every model, while self-rating's *data quality* varied by model (see §5).
- **The representation has real fine-grained structure, not just coarse categories**, in two of the three models (7B and Llama-3B): S5 separability errors land precisely on the

deliberately designed confusable "surprise" pairs, not randomly. This did not replicate cleanly in the 1.5B model, whose confusions were more generic/diffuse.

One result is new and only visible because three models were tested, not two: the correspondence does not peak at the same relative depth across architectures. Both Qwen models peak at the deepest tested layer (~75% depth). Llama-3.2-3B peaks at the middle layer (~46% depth) and is slightly *lower* at the deepest layer.

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4. Interpretation

A language model's self-reported persona identity is not causally disconnected from its internal representational state — the two carry shared, recoverable structure, and this is a general property of instruction-tuned transformer LMs rather than an artifact of one model's training. The cross-family replication (Qwen → Llama) is what licenses the word "general" here; a single-family result would leave open the possibility that this is a Qwen-specific training quirk.

The ceiling-normalized comparison is the more honest basis for comparing magnitude across models: once normalized, self-description correspondence sits in a comparable 0.72–0.93 band across all three models and all three layers, rather than looking like it degrades with smaller model size the way raw RSA alone would suggest. Smaller or differently-trained models don't show a weaker effect — they show a noisier *measurement* of a comparably strong effect, which is a more defensible claim than "bigger models have stronger self-knowledge."

The layer-depth divergence is better read as an open question than as noise: three layers is not a full sweep, so the true Llama peak is not pinpointed, but the direction (meaningfully lower at the deepest layer than at the middle one, consistently across both self-report channels) is too large and too consistent to be a random fluctuation of the permutation test.

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5. Hypotheses this motivates (for the causal follow-up)

1. **H1 — Generality.** The correspondence is a general property of instruction-tuned transformer LMs, not specific to one family or scale. *(Supported, with appropriate caution — n=3 models, 2 families, is strong initial evidence, not an exhaustive demonstration.)*
2. **H2 — Architecture-dependent locus.** The depth at which the correspondence is strongest depends on architecture or training recipe, not a fixed universal fraction of total depth — directly motivating the causal follow-up to test the intervention layer per-model rather than assume it transfers from one architecture to another.
3. **H3 — Channel robustness asymmetry.** Free-text self-description is more robust across model scale/family than forced single-token self-rating, plausibly because free text sits closer to a model's natural output distribution, while terse digit-only answers are more

sensitive to model-specific instruction-following calibration under that model's own chat-template conventions.

4. **H4 — Training recipe over raw scale.** Fine-grained persona-representation structure tracks something closer to training quality/recipe than raw parameter count — the 3B Llama model showed cleaner S5 structure than the 1.5B Qwen model despite being closer in size to it than to the 7B.
5. **H5 — Per-model causal targeting.** Since both magnitude (ceiling-corrected) and layer-location of the correlational effect differ across architecture, planned causal intervention work should not assume a single layer discovered in one model transfers to another.

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6. Rigor and controls — why this research holds up

Three things, taken together, are what make this more than three isolated correlational measurements:

First, every control addresses a specific, nameable alternative explanation, not rigor for its own sake. Partial-RSA rules out "it's just differently-worded prompts." The shuffled-label and permutation-test controls rule out "the RSA pipeline is inflated by an implementation artifact." The designed confusable-pair personas turn S5 separability from a coarse sanity check into a positive control for fine-grained structure — a model that only separated *obviously* different personas would tell you much less than one whose errors land exactly on the pairs built to be hard. The split-half noise ceiling turns raw cross-model RSA comparison, which would otherwise be comparing apples whose sweetness depends on an unknown ceiling, into a comparison that accounts for each model's own measurement reliability.

Second, the extraction mechanics were verified, not assumed, every single time. Three separate times (once per model), the hook-output type, the no-op behavior at alpha=0, the exact position targeted by the hook, the digit-token identities, and the chat-template's generation point were all independently reverified against the live model rather than copied from the previous run. This is what caught the Experiment 2 layer-mismatch before it entered a results table, and what correctly diagnosed Experiment 3's low-confidence ratings as a genuine model behavior rather than quietly accepting or silently patching around it.

Third, cross-family replication is itself the strongest available control. A correspondence that only appeared in one model's pipeline could be explained by something specific to that model's tokenizer, template, or training — a coincidence of this particular RSA computation against this particular architecture. The same qualitative result, surviving independent re-verification of every tokenizer- and template-dependent mechanic, appearing in a second model from an entirely different, independently-trained family, is much harder to explain as anything other than a real property of the phenomenon under study.

What this buys the paper, concretely: a correlational claim that can honestly say it was pre-registered at the layer level, independently re-verified at the mechanics level per model, controlled for the most obvious confound (text wording), checked against a proper statistical null, corroborated by two independently-constructed self-report channels, and replicated across architecture — while still being exactly what it claims to be: correlational evidence, nothing causal asserted or implied.

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7. Limitations and caveats, compiled honestly

Shared across all three experiments:

- **Correlational only.** No causal claims are made or supported — this is explicitly scoped as the correlational motivation study for a separate, future causal (steering/patching) paper.
- **Small, fixed persona set (K=16), two variants per persona, one elicitation-prompt family per channel type.** Robustness to substantially different phrasings, formats, or a larger/different persona taxonomy is untested.
- **Only three layers tested per model**, not a full depth sweep — the exact peak layer (especially Llama's) is located to the nearest of three sampled points, not pinpointed.
- `PRIMARY_LAYERS` **was hardcoded to match the original 7B run's indices** for comparability, not re-derived independently per model — a deliberate, defensible choice, but one that means the specific depth tested for Llama was asserted for layer-count-matching purposes, not independently validated as that architecture's "true" relevant depth.
- **Single run per model, no repeated-seed replication.** Run-to-run stochastic variation (sampled generation, judge-scoring variance) is mitigated by n=16 samples per persona but not separately quantified.
- **S2 judge pass-rate systematically under-rewards quieter, professionally-registered personas** (customer_service, librarian, diplomat, interrogator) across all three experiments — a known, explained judge-calibration limitation, not evidence those personas are poorly expressed (manual eyeballing confirmed in-character text for all of them in every experiment).

Model-specific:

- **1.5B:** self-rating EV calibration notably worse than 7B's; S5 confusions diffuse rather than landing on designed pairs — genuine signal-quality degradation at this scale.
- **Llama-3B:** 66% of self-ratings below the confidence threshold, traced to a real terse-instruction-calibration weakness specific to this model under its own chat template — self-rating RSA for this model should be read with more caution than its self-description RSA, or than Experiment 1's channel-agreement figure. Its chat template also injects a

fixed date-preamble into every system prompt (absent in Qwen), assessed low-risk but not independently proven to have zero effect.

What this does not yet establish: a fully general "holds for all LLMs" claim (three models, two families, is strong initial evidence of generality, not an exhaustive demonstration); or which layer is causally responsible for anything, which is explicitly the next paper's question.

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8. Related work — how this stands in the space

The positioning below is drawn from the project's own literature synthesis conducted ahead of the full Phantasos design (the causal follow-up this correlational study motivates), not from independent re-verification of each cited paper here — treat citation specifics as the project's synthesis, to be checked against the original sources when writing the paper's actual Related Work section.

The reviewed literature clusters into three groups relevant to this result:

- **Generic-concept representational studies** (e.g. work on activation patching/steering around injected concepts like "apple" or "bread," and closed-set forced-choice characterization methodology) establish that models' internal states can be probed and manipulated for arbitrary injected content, and motivate methodological choices used here (closed-set/forced-digit rating over open-ended naming) — but none test identity or persona specifically, and none pair a representational-correspondence test with self-report the way this project does.
- **Self-report validity and persona-geometry work** (demographic/survey ground-truth studies; persona-vector construction papers; self-report-vs-behavior or self-report-vs-generic-agent comparisons) each touch one side of the question this project asks — either persona geometry *or* self-report fidelity — without testing whether a model's self-described identity actually corresponds to its own activation geometry, and without a cross-model replication check.
- **Scalar emotive-state coupling work** demonstrates models' internal states track simple affective scalars, which is a narrower claim than the relational RDM-geometry correspondence tested here across a full K=16 persona set.

What this project adds, specifically: a representational-correspondence test (RSA between activation geometry and self-report) run on identity/persona content specifically — not a generic injected concept — using two independently-constructed self-report channels, controlled for the text-wording confound via partial RSA, and replicated across two independently-trained model families and a 1.5B–7B parameter range. The cross-model replication in particular is not something a single-model correlational study could claim, and it is what turns "we observed a correspondence" into "we observed a correspondence that is not an artifact of one model's training." This correlational result is deliberately positioned as *prior*

motivating evidence for the project's planned causal work (steering/patching, detection/characterization), in the same way the field's generic-concept causal-intervention studies motivated their own closed-set/forced-choice and surface-form-controlled methodology choices before running their causal experiments — this project follows that same discipline, establishing the correlational case first and explicitly deferring causal language to the follow-up study.

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9. What this research finally concludes

Across three models — Qwen2.5-7B-Instruct, Qwen2.5-1.5B-Instruct, and Llama-3.2-3B-Instruct, spanning two independently-trained model families and a parameter range from 1.5B to 7B — a language model's self-described persona identity shows a strong, statistically robust correspondence with the geometry of its own internal activations. This correspondence survives controlling for the confound of prompt-text wording, is corroborated by two independently-built self-report channels, and is not explained away by chance or by one model's particular reliability profile.

The effect is general but not uniform: its magnitude is consistent across models once corrected for reliability, but its location — which layer carries it most strongly — is not fixed across architecture, peaking deepest in both Qwen models and at mid-depth in Llama. Free-text self-description is the more dependable self-report channel across model family and scale; forced single-digit self-rating, while significant everywhere, is more sensitive to model-specific instruction-following calibration and should be weighted accordingly.

This is, deliberately, a correlational result — the paper's own scope commits to stopping here. What it establishes is exactly what was intended: good, now cross-model-replicated evidence that the correspondence is a real and general phenomenon, substantial enough to motivate — and specific enough in its layer-depth findings to help design — the causal intervention study that is this project's natural next step.